Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 165 (2019) 196–208

Procedia Computer Science

www.elsevier.com/locate/procedia

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019, ICRTAC 2019

Certain Investigation on MANET Security with Routing and Blackhole Attacks Detection

Jaikumar Vinayagam [a,*], CH. Balaswamy[b], K. Soundararajan[c]

[a]Research scholar, JNTUA, Anathapuramu, A.P., India
.[b]Professor, Gudlavalleru Engg. College, A.P.,India
[c]Rtd Senior Professor, JNTUA, Anathapuramu, A.P., India

Abstract

Mobile Ad-hoc Network (MANET) is a self-configuring network depending upon infrastructure, also is a significant technology which supplies virtual hardware and software resources as per requirement of mobile ad hoc network. The Intrusion Detection System (IDS) has been established to protect Cross Interior (routing) at source to goal from different types of attack. Current IDS approaches can be classified as either anomaly or signature-based methods such as Rule-based expert systems, State transition analysis, Bayesian alarm networks aspects are various challenges with respect to the efficiency as well as performance. The major challenge is the compromise of performance because of unavailable resources with respect to the MANET. To increase the MANET environment's performance, various techniques are employed for routing, security purpose. An efficient security module requires a routing in terms of packet loss. The Integrated Cross Interior (ICI) structure for Intrusion Detection System (IDS) namely ICIs for IDS are proposed for exactness of black hole attacks detection. The ICIs for IDS performs the task of routing of the mobile nodes. The proposed (ICIs for IDS) produces the enhancement of Intrusion Detection System (IDS) with routing algorithm named as Integrated Cross Interior algorithm that performs maps the security with the secure IDS communication and distributes the packets among different destination, based on priority. This paper proposes a novel Integrated Cross Interior structure for Intrusion Detection System (ICIS for IDS) to secure our network from black hole attacks. This algorithm is proposed for node routing and node security by considering maximum throughput 83.5% with minimum routing cost, response time with 6.3ms. The experimental results specify that ICIs for IDS based security policy effectively minimizes the response time as well as the mobile routing cost, response time of an application with respect to other existing approaches such as IDS with AODV. The proposed algorithm is experimented using Network Simulator-2 and the results demonstrated that the performance of the algorithm is better than other conventional algorithms like IDS with AODV.

* Corresponding author. Tel.: +919866401062.
*E-mail address:*vinayagam.jai81@gmail.com

1877-0509
10.1016/j.procs.2020.01.091

Keywords: Self-Configuring Networks; Virtual Hardware; Routing; Security Policy; Cost; Response Time; Mobile Makespan; Routing Cost; Throughput;

1. Introduction

MANET are come about always in the form of connection-less, permanent centralized server-less-network model for dynamically stipulation and access the data as a remote controlled and scalable mobile node as well as routing services through efficiency routing security techniques. In routing path, the routing security techniques are works based on the data blockage and the communication time [8]. Scalable mobile nodes discovery is a significant technology behind Mobile Ad-hoc Network which allows sharing of RREQ/RREP (Route Request/Route Reply) among multiple nodes for the utilization of the source and destination to the complete communication toward the route. Node routing consists of attackers that provide a separated attacked environment with respect to the malicious nodes [6]. The entire functionality of Mobile ad-hoc Network can be categorized into three significant modules namely 1. Node Formation, 2. Routing and 3. Security.

Node Formation module is the process in which the source requested RREQ/RREP is send over the nearby nodes for the availability. Once the provisioning target node is replied, then the next step is data routing. For the enhancement of Intrusion Detection System (IDS) performance, both modules such as Node Formation, Routing should be given equal importance so that it allows the destination to send the required data optimally as per the nodes routing security [10].

Nodes security module, the RREQ/RREP packets are send to the destination, subsequently target node is replied hence the node is approved for data transmission over these two nodes without any interruption like black hole attack in MANET. This data communication is called security module. The nodes security module targets the destination and route pattern over it for attacker as per the routing cost, response time of the Node Communication and Attacker Involvement. It basically balances the node communication and attacker involvement of the nodes security module in order to minimize the routing cost, and enhances the performance by decreasing the response time. Several challenges are encountered in the routing nodes security environment. Out of the numerous experiments faced in the security threads environment, one of the critical challenges is to accomplish the attacker free transmission. i.e., maximize the node energy. The node energy maximization is a process of removal of attacker from the alive network by reducing the attacker of the node, so as to maximize the node energy; we can reduce the QoS based security policy like Node energy, node lifetime of attacker [4].

In the Integrated Cross Interior algorithm, the security enhances set amongst participating nodes and the process is performed with the help of various algorithms such as QoS based security, Integrated Cross Interior algorithm and secure IDS communication. This process is recognized in the following fig 1.

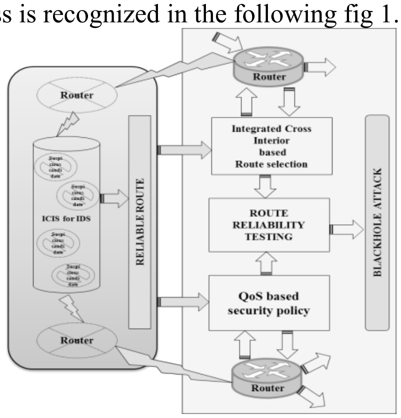

Fig 1. MANET Security with Routing and Blackhole Attacks Detection

In the above Fig 1 discussed about the MANET security with routing and blackhole attacks detection. In this process, QoS based security policy techniques are discussed, this technique named as Integrated Cross Interior algorithm. The IDS communication worked based on Integrated Cross Interior algorithm. Integrated Cross Interior algorithm process used to perform the task of routing of the mobile nodes for routing cost, response time. Finally, security with the secure IDS communication and distributes the packets to improve the routing, security by maximize throughput and minimize routing cost. To minimize the security threads in the area of LAN, WAN, WLAN against Intrusion Detection System (IDS), this proposed system can provide strong techniques to distinguish unauthorized accomplishments in the MANET [10].

2. QoS Support in MANETs

Li et al (2014) proposed the new routing protocol for hybrid wireless networks. Li et al (2014) were trust different network namely, MANET and wireless medium [1]. Their research discussed about the Distributed routing protocol QoS such as mobility-resilience, overhead, scalability, and transmission delay for resolve the resource scheduling problem in the neighbor selection with respect to the transmission delay and time. This will help to increase throughput and data redundancy. To manage the Li et al (2014) process, Bouk et al (2012) introduced the QoS and non-QoS Path Parameters between the Mobile Node and Gateway Node for gateway selection [2]. Compare to the Li et al (2014) model, Bouk et al (2012)model focused on the end-to-end delay, link capacity, node speed, number of hops, path availability period and residual energy for updated path. Even though Hanzo et al (2011) is overcome the Li et al (2014) model and Bouk et al (2012) model by routing in mobile ad hoc network with respect to the QoS assurances. Hanzo et al (2011) model deals two protocols such as QoS-aware routing protocol and admission control protocol [3] [11]. Hanzo et al (2011) model used to improving the performance their protocols in terms of mobility, routing reliability shadowing, throughput, transmission efficiency, varying link [8].

3. QOS Based Security Policy

The Intrusion Detection System (IDS) routing, workings based on the Integrated Cross Interior algorithm. It was developed and analysed for the purpose of routing cost, response time in the MANET. Integrated Cross Interior algorithm is a simple algorithm, used for security with the secure IDS communication and distributes the packets [10]. Integrated Cross Interior algorithm was implemented and combines with Past Interaction History Based nearby Path Secure Routing Algorithm. Also simulated and compared on various metrics such as Attacker Detection Ratio, Cluster Distance (CE Distance), Overhead measure, Throughput measure.

Integrated Cross Interior algorithm solves the following problems:

- Integrated Cross Interior algorithm mainly solves the data communication process deficiency around most IDS system.
- By using of path origin selection, it resolves the problem occurs in shortest route formulation.
- Integrated Cross Interior algorithm solves separation/travel time problem. It causes the network route options. This is preventing the block-hole attack, also travel time and node location too.
- Integrated Cross Interior algorithm solves route formation (routing) data transparent rate
- Integrated Cross Interior algorithm rectifies these concerns against optimal routes which are also saved in Past Interaction History also provide the different shortest path which does not consider any block-hole attack.

Integrated Cross Interior algorithm can be improved so as to enhance its performance, if following security and routing issues are focused:

- Integrated Cross Interior algorithm explains the scenario of routing of one sender node to another receiver node. Perhaps, the concept of security and routing can also be added to it that leads to improve the overall performance of the MANET Communication. Hence the discovery, node communication and packet forecast increases.

In addition to this, an effort can be made to automate the Integrated Cross Interior algorithm, in order to develop organize MANET environment.

Meanwhile, it is QoS based security policy focus on above stated issues; another Intrusion Detection System i.e. integrated Cross Interior (ICI) Structure for Intrusion Detection System (IDS). These ICIs for IDS works based on the Cross Centric Attacks Framework. This framework covers Preset Time Interval Algorithm (PTIA). In the proposed ICIs for IDS, resolves block-hole attack by broadcast the messages over the nearby neighbor node. This message broadcasting is used to find flagging nodes between two participating nodes. In this era, the time period and packet delivery ratio are chosen for individuals' conversation [7]. This time period and packet delivery ratio is integrated with Intrusion Detection System (IDS) which is based on the Preset Time Interval Algorithm (PTIA) of the respective cluster nodes.

4. Integrated Cross Interior (ICI) Structure for Intrusion Detection System (IDS)

ICIs for IDS mimic the Intrusion Detection System activity of the participating nodes on the IDS communication system. In MANET, the source node acts as the routing communication initiator and need destination i.e. receiver node. From this source node and receiver node, we can calculate the communication efficiency of each exchange with respect to the trusted node as well as the attacker nodes [9].

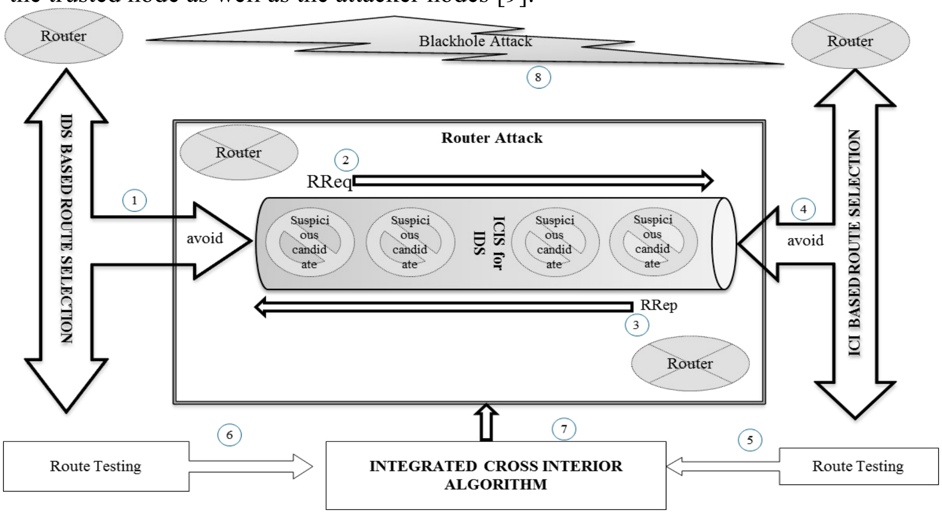

Fig 2. Processing steps for Integrated Cross Interior (ICI) Structure for Intrusion Detection System (IDS)

The above Fig 2 discussed about the Processing steps for Integrated Cross Interior (ICI) Structure for Intrusion Detection System (IDS). It contains the following processing steps,

Step 1: Route selection is done on the Intrusion Detection System (IDS) by avoiding the Suspicious candidates on the router attack process

Step 2: Route request sent over the nearby nodes for getting the exact nearby node selection

Step 3: Route request received from the nearby nodes for confirming the exact nearby route selection

Step 4: Route attacker confirmation is done by the Integrated Cross Interior (ICI) by avoiding the Suspicious candidates

Step 5: Router Testing Module are applied to the Integrated Cross Interior, like Static methods.

Step 6: Router Testing Module are applied to the Integrated Cross Interior, like dynamic method.

Step 7: The tested Route must have integrated with the Integrated Cross Interior (ICI) Structure for Intrusion

Detection System (IDS)

Step 8: After a specified period of time, blackhole attack might be detected with distrustful and unwilling specific behaviour.

4.1 Response Time Assignment

For response time, the routes are arranged according to their network size of the MANET. This response time assignment considers three part of the process flow. On the other hand, the higher routing efficiency [9] with higher response time causes the data transaction by.

i. The higher value if fixed time interval $T_{(Src,Revr)}$ is greater than response time $PRN_{(Src,Revr)}$, the process data transaction will get maximum communication response time efficiency
ii. Meanwhile remaining fixed time interval were processes for next transaction.
iii. Calculate total energy of the transaction at iteration of response time efficiency of both trusted and attacker node.

According to this, such total maximum communication response time efficiency of trusted node is computed with respect to the remaining energy.

Algorithm for Integrated Cross Interior algorithm

ICIa←StartPkt(pkt)
T_{delay}← ICIa _STREAM(pkt)
$T_{arrival}$←Max_Route[$T_{delay(Normal)}$, $T_{delay(RRED)}$])
If pkt.arrival_time is within [$T_{arrival}$, ($T_{arrival}$ +$T_{fixed\ time}$)]
then
find route as valid packet with valid node
Finally node is recorded corresponding to the shortest path
Else
node re_transmit the node's packet again to calculate the delay time.
STREAM [Max_Route].CA_NODE ← maximum of retransmit the node's packet of Route
if STREAM[Max_Route]. CA_NODE >=0
then
ICIa_Transfer(pkt)→ Next Neighbour_Node
If ICIa_Transfer drops pkt
then
ICIa_NOde←!(pkt.arrivaltime)
MIN_Route(pkt)←NODE($T_{arrival}$== ($T_{arrival}$ +$T_{fixed\ time}$))
Else
STREAM[Max_Route]. $T_{arrival}$ ← ICIa_Node
drop.(Start_pkt)

$$\text{Avg packet size to send individual node} = \text{Node}_{\text{before send packet}} * \left(1 - 2^{-\text{no.of nodes (snd)}}\right) + \text{Node}_{\text{after send packet}} * \left(2^{-\text{no.of.nodes (rcvd)}}\right) \quad (1)$$

$$\text{Transmission Efficiency Ratio} = \sum_{T_{st}}^{T_{FxT}} \left\{ \frac{\text{Tot}_{\text{Pkt (Rcv)}}}{\text{Tot}_{\text{Pkt (snd)}}} \middle| \text{Total No. Nodes in Group}(T_n) \right\} \quad (2)$$

$$\text{TER} = \sum_{T_{st}}^{T_{FxT}} \left\{ \frac{\text{Tot}_{\text{Pkt(Rcv)}}}{\text{Tot}_{\text{Pkt(snd)}}} \right\} \left(\bigcup_{T_{\text{Fixed time}}} \text{Group\_Node}(T_{I(\text{pkt})}) \middle| \bigcup_{T_{\text{Fixed time}}} \text{Neighbour\_Node}(T_{I(\text{pkt})}) \right) \quad (3)$$

$$\text{Transmission Delay (TD)} = \sum_{T_{\text{start time}}}^{T_{\text{Fixed Time}}} \left\{ \frac{(A_{\text{pkt (t)}} - T_{FxT}) + (S_{\text{pkt (k)}} - T_{FxT})}{T_n} \right\} \quad (4)$$

$$\text{Target Utilization Index (TUI)} = \left\{\text{Tot}_{\text{Pkt (snd)}} \middle| \text{Tot}_{\text{Pkt (Rcv)}}\right\} \quad (5)$$

$$\text{Node Utilization Index (NUI)}_{\text{Fixedtime}} = \oint_{\text{Initial routepath}}^{\text{finalroute path}} \frac{(\text{Time taken for one pkt} + \text{path disdance})}{\text{T}_{\text{Fixed Time}}} \quad (6)$$

Return MIN_Route(pkt) replace the following abbreviation everywhere:
$\text{Fixed Time (delay)} = \text{T}_{\text{FxT}}$, $\text{Starting Time} = \text{T}_{\text{st}}$, $\text{Total. No. of. Packet Received} = \text{Tot}_{\text{Pkt (Rcv)}}$, $\text{Total. No. of. Packet send} = \text{Tot}_{\text{Pkt (snd)}}$, $\text{Time(individual}_{\text{Node}}) = \text{T}_{\text{I(Pkt)}}$, $\text{Arrival time for single Packet} = \text{A}_{\text{pkt (t)}}$, $\text{send time for single Packet} = \text{S}_{\text{pkt (k)}}$ and $\text{total No. Nodes in Group} = \text{T}_{\text{n}}$

Algorithm for Route Discovery:

Def:NetworkSize(No.of.Nodes=N_n, Pkt Life Time=P_{lt},MalNode=A_n , RouteLink=L_r, PixedLife Time =P_{lx}):

$$\text{RouteLink(Lr)} = \left\langle \left(\frac{\text{Normal Packet Life Time}}{\underbrace{\text{pixed time inteval}}_{\text{Normal Node}}} * 2 \right) + 1 \middle| \left(\frac{\text{Malicious Node Life Time}}{\underbrace{\text{pixed time inteval}}_{\text{Malicious nodes}}} * 2 \right) - 1 \right\rangle \quad (7)$$

$$\text{NetworkSize} = \left(\text{fixed time interval} * \left(\frac{\text{Normal Nodes Life time}}{2} + \frac{\text{Malicious Node Life time}}{2} \right) \right) \quad (8)$$

$$\text{NodeRoute} = (\text{Network size}) \left\| \left(\frac{\text{Normal Node Lifetime}}{2} \middle| \frac{\text{Malicious Node Life time}}{2} \right) \right\| \quad (9)$$

```
MinRoute=InitialModetime interval (StartNode,NeighbourNode);
MinRoute[StartNode:Null]= Z[CA_Node:Head]= 1;
MinRoute[Null:NextStartNode]=[Head:BroadcastNode] = 1;
If((MinRoute(CA_Node)&&MinRoute(BroadcastNode))
        for Broadcast in range(density);
MinRoute(CA_Node, BroadcastNode)= rand(VlaidNodeRoute) * 2
MinRoute [BroadcacstNode, NeighbourNode] = 1;
forNeighbourNoderange(Next NeighbourNode):
neighbours = []
                if start Node > 1:
neighbours.append ((neibourNode,startNode- 2))
        ifstartNode<FixedlifeTime-2:  neighbours.append((neibourNode, startNode+ 2))
                ifNeighbourNode> 1: neighbours.append(FirstNeibourNode-2,StartNode))
                ifTotalCount (NetworkSize):
Min(RoutePath)=neighbours[rand(Start,
TotalCount(NetworkSize - 1)
```

Whenever the node is in critical path, the Min_Route Path may be calculated by using the above route discovery algorithm. This describes, the nodes want to communicate with other nodes, and the packet messages are spread over the nearby neighbor nodes because of finding the next nearest node. Whenever, the route is complicated to communicate with others, this route discovery algorithm, the packet messages are spread over the nearby neighbor nodes. To find correct nearest path, the route discovery algorithm with that protocol patterns are delimited with each other and forms the finest route [12]. It is used to increase packet delivery ratio of the individual node and also reduce Packet delay for those routes. Once a route of the node is established without any blackhole attacks, the destination node path are helps to deliver the packet for next sequential node.

5. Benchmark functions

A set of 500 nodes are participated to evaluate the performance of QoS based security policy. Section 3 and 4 presents the functions used in the proposed research. The notation table for Network Simulator-2 were given in the following table.

Table 1. Notation Table

mn	Mobile node
Pkt	Packet
T_{delay}	Common processing Time delay
$T_{arrival}$	Arrival time delay
T_{st}	Starting Time
T_{FxT}	Fixed time interval
TER	Transmission Efficiency Ratio
TD	Transmission Delay
TUI	Target Utilization Index
NUI	Node Utilization Index
$Tot_{Pkt\,(Rcv)}$	Total Number of Packet Received
$Tot_{Pkt\,(snd)}$	Total Number of Packet send
$T_{I(Pkt)}$	Time taken for individual Node selection
$A_{pkt\,(t)}$	Arrival time for single Packet
$S_{pkt\,(k)}$	sending time for single Packet
T_n	Total Number of Nodes in Group
P_{lt}	Packet Life Time
N_n	Number of Nodes
A_n	Malicious Node
L_r	Route Link
(Lr)	Route Link
CA	Cluster Agent

The optimization performance evaluation [5] of QoS based security policy is done with possible combinations of various routing paths, for selective paths are chosen from the widely adopted network. The security analysis of the Integrated Cross Interior algorithm and Preset Time Interval Algorithm (PTIA) are performed. It shows that Integrated Cross Interior algorithm and preset time interval algorithm (PTIA) to removing the black hole attack and also both algorithms monitor the overall functioning of the MANET by removing the malicious activities and hence out perform in comparison with other algorithms.

The performance of QoS based security policy is evaluated through extensive simulation experiments conducted on the widely used Network Simulator-2. Following performance metrics are used to evaluate the performance of QoS based security policy parameters:

Table 2. Transmission Efficiency Ratio

Attacker Test	**Attacker Training**	**Attacker Confirm**	**Transmission Efficiency Ratio (TER) (%)**		
Suspect Phase 1	**Suspect Phase 2**	**Number of attackers**	**IDS with AODV**	**ICIs for IDS with attack**	**ICIs for IDS without attack**
12	07	05	89.50	91.90	95.50
14	04	10	81.90	86.63	87.20
18	03	15	78.00	80.60	83.00
34	14	20	72.40	76.50	79.00

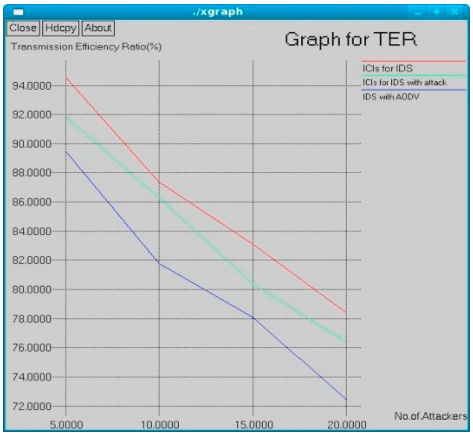

Fig 3. Transmission Efficiency Ratio

Fig 3 give representation of Transmission Efficiency Ratio, ICIs for IDS without attack is unable to establish after a calculation which network is not fake or counterfeit, and so we take to be the case it merely serve as the medium for transmission the data between hops. Blackhole attack made less in size, amount or degree whenever one or both of ICIs for IDS have not spread origin authentication. The comparative results on three quality domain attacks namely IDS with AODV ICIs for IDS with attack ICIs for IDS without attack shows when make work for a particular purpose ICIs for IDS the following inference were taken. ICIs for IDS without attack take the efficient Transmission Efficiency Ratio in terms of number of attackers' comparison to other combination. In terms of number of attackers and Transmission Efficiency Ratio, the proposed ICIs for IDS return the best result by 95.50%. Performance metrics memory utilization was comparatively less (10MB), and execution time and efficient (90.19%) in ICIs for IDS [8] [9].

Table 3. Transmission Delay

Attacker Test	**Attacker Training**	**Attacker Confirm**	**Transmission Delay (TD) (ms)**		
Suspect Phase 1	**Suspect Phase 2**	**Number of attackers**	**IDS with AODV**	**ICIs for IDS with attack**	**ICIs for IDS without attack**
12	07	05	5.25	3.50	0.95
14	04	10	12.10	9.25	7.25
18	03	15	22.00	19.25	13.25
34	14	20	28.60	25.00	18.40

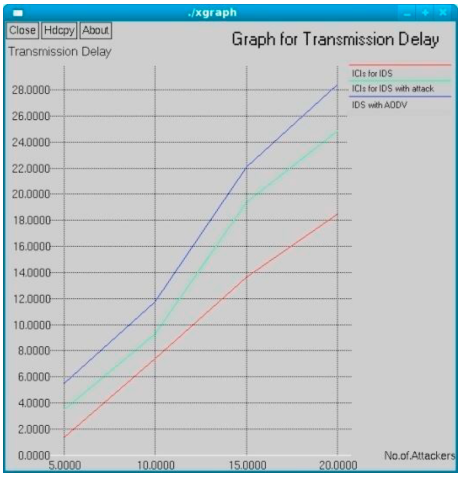

Fig 4. Transmission Delay

Fig 4 give a description of two suspect phase (Suspect Phase 1 and Suspect Phase 2) obtainable for attacker confirmation. Suspect Phase 1= {12, 14, 18 and 34} attackers are tested on the various situation and Suspect Phase 1= {07, 04, 03 and 14} attackers are reduced respectively. The relative results on three different domain attacks namely IDS with AODV ICIs for IDS with attack ICIs for IDS without attack shows when we applying ICIs for IDS the following inference were taken. ICIs for IDS without attack take the efficient for Transmission Delay (TD) compare to other IDS with AODV and ICIs for IDS with attack. The result of performance metrics like memory utilization in ICIs for IDS without attack were comparatively less (8MB), and in terms of execution time is efficient (69.68%) on ICIs for IDS with Transmission Delay based Attacker Confirmation and its delay.

Table 4. Transmission Throughput

Attacker Test	**Attacker Training**	**Attacker Confirm**	**Transmission Throughput (TT)**		
Suspect Phase 1	**Suspect Phase 2**	**Number of attackers**	**IDS with AODV**	**ICIs for IDS with attack**	**ICIs for IDS without attack**
12	07	05	9.00	13.00	14.75
14	04	10	11.90	16.00	17.25
18	03	15	12.10	21.80	23.00
34	14	20	14.50	27.00	28.50

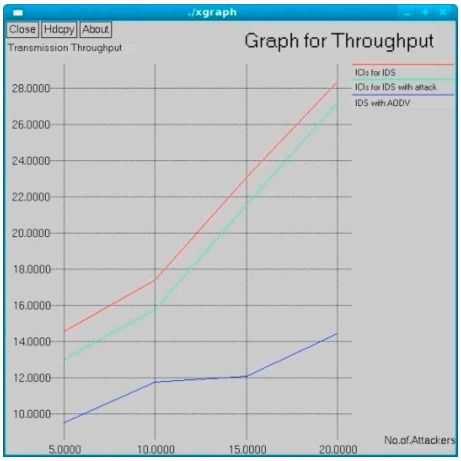

Fig 5. Transmission Throughput

Fig 5 also give a description of two suspect phases (Suspect Phase 1 and Suspect Phase 2) available for Attacker Confirmation. The comparative results on three different combination i.e. IDS with AODV, ICIs for IDS with attack, ICIs for IDS without attack, the ICIs for IDS without attack is efficient minimum throughput (14.75%) and maximum throughput 28.50%) is best in ICIs for IDS compare to other. The performance metrics like memory in 20News group dataset were comparatively less (7MB) than IDS with AODV and in terms of execution time is ICIs for IDS were 10.71% faster.

Table 5. Target Utilization Index

Attacker Test	**Attacker Training**	**Attacker Confirm**	**Target Utilization Index (TUI)**		
Suspect Phase 1	**Suspect Phase 2**	**Number of attackers**	**IDS with AODV**	**ICIs for IDS with attack**	**ICIs for IDS without attack**
12	07	05	9.00	5.00	2.75
14	04	10	13.25	9.10	8.75
18	03	15	18.25	15.00	13.10
34	14	20	21.00	17.10	15.00

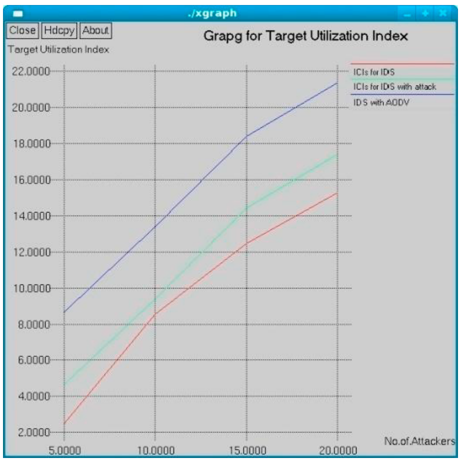

Fig 6. Target Utilization Index

Fig 6 describes, there are two suspect phase (Suspect Phase 1 and Suspect Phase 2) available for Attacker Confirmation. Suspect Phase 1= {12, 14, 18 and 34} attackers are tested on the various situation and Suspect Phase 1= {07, 04, 03 and 14} attackers are reduced respectively. The results on three different combination IDS with AODV, ICIs for IDS with attack, ICIs for IDS without attack shows, when we applying proposed ICIs for IDS algorithm, the prior conclusions on the basis of direct observation were taken and ICIs for IDS produces the best minimum Target Utilization Index is 2.75% and maximum Target Utilization Index is 15.00% is best in ICIs for IDS without attack. The performance metrics like memory in Target Utilization Index were comparatively less (10MB). In terms of execution time in ICIs for IDS were 96.21%.

Table 6. Node Utilization Index

Attacker Test	**Attacker Training**	**Attacker Confirm**	**Node Utilization Index (NUI)**		
Suspect Phase 1	**Suspect Phase 2**	**Number of attackers**	**IDS with AODV**	**ICIs for IDS with attack**	**ICIs for IDS without attack**
12	07	05	1.00	5.00	8.00
14	04	10	2.50	7.55	12.00
18	03	15	5.00	8.75	17.00
34	14	20	7.00	9.00	23.00

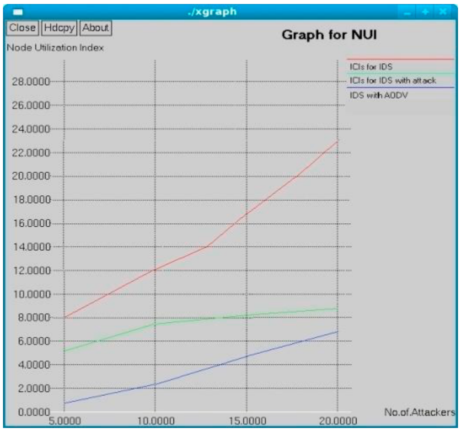

Fig 7. Node Utilization Index

Fig 7 describes, there are two suspect phase (Suspect Phase 1 and Suspect Phase 2) available for Attacker Confirmation. Suspect Phase 1= {12, 14, 18 and 34} attackers are tested on the various situation and Suspect Phase 1= {07, 04, 03 and 14} attackers are reduced respectively. The results on three different combination IDS with AODV, ICIs for IDS with attack, ICIs for IDS without attack shows, when we applying proposed ICIs for IDS algorithm, the prior conclusions on the basis of direct observation were taken and ICIs for IDS produces the best minimum Target Utilization Index is 2.75% and maximum Target Utilization Index is 15.00% is best in ICIs for IDS without attack. The performance metrics like memory in Target Utilization Index were comparatively less (7MB. and in terms of execution time in ICIs for IDS were 91.62%.

Following simulation settings are used in the experiments:

Table 7. Simulation Parameter

Simulation Parameter	Value
Channel & capacity	Wireless mobile Channel with 2Mbps
Transmit power	0.3J
Receiver power	0.1J
Initial energy	11J
Antenna	Single Omni Antenna
Max packet	500
Number of nodes simulated	100
Simulation time	200 ns
Routing	Proposed Routing Model
Initial Routing Node count	100
Node Pattern	Trust Value (Sender, Receiver)
Time interval	0.20 ns ~ 0.45 ns
End to end delay	0 ns ~ 0.0.15 ns

Table 7 Shows, Simulation parameters that the proposed rerouting technique, it is used to identify the system environments of Network simulator 2.

6. Conclusion

Makespan of Routing and Security in Intrusion Detection System (IDS) is novel security outlines that keep away the black hole attack from MANET. This paper firstly focused on secure routing to a destination on opened network and also ensures the convenient and reliable communication with QoS Parameters. This proposed research the addresses authenticity services and security dimensions in MANETs. This proposed system was fulfilled QoS based security policy such as Communication Efficiency of Routing Cost Assignment and Communication Efficiency of Response Time Assignment. More over the security policy narrative is intended to provide a secure environment between routing cost assignment and response time assignment. However, it extends protection to routing services with high-cost security. It also provides a Benchmark functions such as Communication Efficiency Ratio, Average Transmission Delay, Average Throughput, Destination Utilization Index, Neighbor Utilization Index, Packet transmission deadline, Packet arrival rate, Attacker Failure Rate. These measures are used to self-estimate the proposed authenticity services and security dimensions in MANETs. Integrated Cross Interior algorithm and Preset Time Interval Algorithm (PTIA) providing security across all MANET's nodes with lower routing cost, response time in their respective routing protocols. In future the various stages of the proposed routing models can be parallelized to improve their efficiency. Further combination of Attacks and associative attacks can be probed and combined with MANET techniques.

References

[1] Meng L, Song W, "Routing protocol based on Grover's searching algorithm for Mobile Ad-hoc Networks." *China Communications* 2013 Mar;10(3):145-56.

[2] Bouk SH, Sasase I, Ahmed SH,"Gateway discovery algorithm based on multiple QoS path parameters between mobile node and gateway node." *journal of communications and networks*. 2012 Aug;14(4):434-42.

[3] Kumar M, Bhandari R, Rupani A, Ansari JH, "Trust-based Performance Evaluation of Routing Protocol Design with Security and QoS over MANET."In2018 International Conference on Advances in Computing and Communication Engineering (ICACCE) 2018 Jun 22 (pp. 139-142). IEEE.

[4] Surendran S, Prakash S,“An ACO look-ahead approach to QOS enabled fault-tolerant routing in MANETs.”', *China Communications*. 2015 Aug;12(8):93-110.

[5] Ch RM. M-LionWhale, “Multi-objective optimization model for secure routing in mobile Ad-hoc network.”, IET Communications. 2018 Mar 21.

[6] Sangeetha V, Kumar SS, “Detection of malicious node in mobile ad-hoc network.”In2018 International Conference on Power, Signals, Control and Computation (EPSCICON) 2018 Jan 6 (pp. 1-3). IEEE.

[7] Khan MS, Midi D, Khan MI, Bertino E, “Fine-Grained Analysis of Packet Loss in MANETs.” IEEE Access. 2017;5:7798-807.

[8] Nguyen DQ, Toulgoat M, Lamont L, “Implementing voice over Internet protocol in mobile ad hoc network–analysing its features regarding efficiency, reliability and security. The Journal of Engineering. 2014 Mar 21;2014(5):184-92.

[9] Sheikh NA, Malik AA, Mahboob A, Nisa K, “Implementing voice over Internet protocol in mobile ad hoc network–analysing its features regarding efficiency, reliability and security.” The Journal of Engineering. 2014 Mar 21;2014(5):184-92.

[10] Nadeem A, Howarth MP, “A survey of MANET intrusion detection & prevention approaches for network layer attacks.” IEEEcommunications surveys & tutorials. 2013 Mar 9;15(4):2027-45.

[11] Jung JY, Choi HH, Lee JR., “Survey of Bio-Inspired Resource Allocation Algorithms and MAC Protocol Design Based on a Bio-Inspired Algorithm for Mobile Ad Hoc Networks. *IEEE Communications* Magazine. 2018 Jan;56(1):119-27.

[12] Chen YH, Hu CC, Wu EH, Chuang SM, Chen GH. A Delay-Sensitive Multicast Protocol for Network Capacity Enhancement in Multirate MANETs. *IEEE Systems* Journal. 2018 Mar;12(1):926-37.